

ERIN WALSHAW

778-681-7171 | erinwalshaw@outlook.com | linkedin.com/in/erinwalshaw | github.com/ewalshaw

EDUCATION

University of Waterloo
Bachelor of Computer Science

Sept. 2024 – August 2029
Waterloo, ON

EXPERIENCE

Manulife | Business Technology Analyst Intern

September – December 2026

- Incoming: will support digital transformation initiatives, working with datasets and relational databases to generate insights and dashboards for technology and business decisions.

Poket | Software Developer Intern

January – April 2026

- Shipped production bug fixes and usability improvements to the Poket Portal in JavaScript, HTML, and CSS
- Migrated the codebase from Angular 16 to 17, resolving breaking API changes and dependency conflicts.
- Instrumented the Poket Portal with GTM/GA4/Microsoft Clarity with 30+ custom events to enable behavioral analysis for engineering prioritization
- Redesigned the client-facing site for SEO and LLM-search discoverability, targeting both traditional rankings and AI-powered search surfaces

University of Waterloo | Machine Learning Research Assistant

September – December 2025

- Implemented causal intervention pipelines in PyTorch to replicate core results from the JAM paper, manipulating LLM latent vectors to steer generative outputs toward target behavioral profiles.
- Built reproducible experiment infrastructure for comparing latent-space steering methods across model checkpoints.
- Formalized model behavioral constraints as geometric structures over latent representation space as part of an alignment research framework.

University of Waterloo Futures Lab | AI + UX Prototyping

September – December 2025

- Built an AI-powered learning tool for hardware design courses using Gemini and Google AI Studio, in an 8-week interdisciplinary program with Master's/PhD students and Google mentors
- Presented prototype and user testing results at the Google-UWaterloo Symposium on the Future of Learning.
- Selected as 1 of 5 students for a panel interview with Google featured on their blog.

PROJECTS

TinyIR | Python, scikit-learn

Repo

- Built a command-line information retrieval engine indexing local text corpora via tokenization, an inverted index, and TF-IDF vectorization, ranking search results by cosine similarity.
- Designed a persistent index format (pickled vectorizer, vocabulary, postings list) enabling repeat queries without re-indexing.
- Covered tokenization, inverted-index postings, TF-IDF shape, and ranking order with a pytest suite; wired into CI via GitHub Actions.

pyglance | Python (AST, dependency-free)

Repo

- Built a dependency-free static analyzer that parses Python source into an AST to detect unused imports, functions exceeding a length threshold, and unreachable code after terminating statements.
- Implemented circular-import detection across local modules via import-graph traversal.
- Recursively analyzes directory trees while skipping virtual environments and build artifacts; exits non-zero on findings for CI use.

Carbon Calories | Python, React, Next.js, MongoDB, Gemini API

- Built a full-stack food-tracking app delivering per-user carbon insights and meal recommendations, integrating a Next.js/React frontend with a Python backend and Gemini API for AI-powered meal analysis.

TECHNICAL SKILLS

Languages: Python, Java, C/C++, JavaScript, TypeScript, SQL

Frameworks/ Tools: React, Next.js, Node.js, Git/GitHub, GitHub Actions

Concepts: Data structures and algorithms, information retrieval (TF-IDF, inverted indexing), static analysis, Unix/Linux

LEADERSHIP

Women in Computer Science Committee | Director of Finance

May – August 2026

- Manage committee budgets, cheque requests, and funding relationships for a 100+ member student org.